Hotel Platform: Developer Guide & Feature Specification

Simplified System Overview, Core Modules, Workflow Fixes, and High-Value Feature Additions

1. Project Overview: What We Are Building

We are building a unified Hotel Guest Experience and Internal Operations Platform. Instead of building a bloated system that tries to replace the hotel's existing Property Management System (PMS) or Point-of-Sale (POS), our app acts as the fast, mobile-first operational layer connecting guests and hotel staff in real time.

The 3 Application Surfaces: The platform has three lightweight web/mobile surfaces connected to one modular backend:

- **1. Guest Mobile Web Portal (No App Download):** A zero-install mobile web portal opened by scanning the in-room QR code or clicking a pre-arrival SMS/WhatsApp link. Guests use it to verify their stay, request room cleaning or towels, order room service, report maintenance issues with a photo, view their live bill, and check out digitally.
- **2. Frontline Staff Mobile App (PWA):** A simple, large-button mobile Progressive Web App (PWA) for Housekeeping, Engineering, and F&B runners. Staff see only their assigned tasks, tap to accept/complete work, follow short room checklists, and update room readiness.
- **3. Supervisor & Front Desk Web Console:** A tablet/desktop dashboard for Supervisors, Front Desk, Duty Managers, and the GM to monitor SLA timers, reassign tasks, approve room blocks or service-recovery comps, complete shift handovers, and reconcile failed payment/PMS syncs.

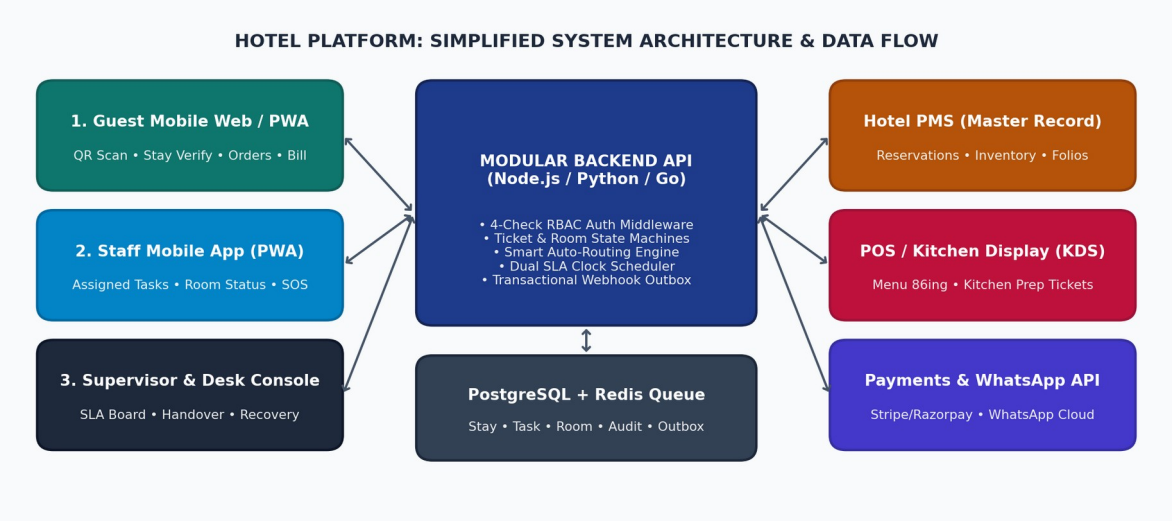


Figure 1: Simplified 3-Tier Architecture — Frontend Surfaces, Modular Backend, and External Systems of Record.

> **SYSTEM-OF-RECORD RULE:** Do NOT rebuild the PMS or Payment Vault. The PMS owns reservations, room inventory, and master folios; the POS owns kitchen tickets; the Payment Gateway owns card tokenization; our app owns the Guest Portal, Task Routing, SLA Timers, and Audit Logs.

2. Four Critical Workflow Fixes to Implement in Backend Logic

Before building individual screens, the engineering team must fix four logic gaps present in the initial flowcharts:

- **Fix 1 — Payment Success != Booking Confirmation:** Never mark a reservation 'Confirmed' solely because the payment gateway returned 200 OK. Maintain separate payment_state (Authorized/Captured) and booking_state (PMS_Pending/PMS_Confirmed). If payment succeeds but the PMS API times out, show 'Pending Confirmation', queue a background retry, and alert the Front Desk reconciliation queue so the guest is never double-charged.
- **Fix 2 — Service Completion Loops Back to Stay Dashboard:** In the initial guest flowchart, completing a service request flowed directly into post-stay checkout. In code, completing a towel or maintenance request must loop back to the Guest Stay Dashboard. Only an explicit, confirmed PMS checkout event triggers final invoicing and post-stay review messages.
- **Fix 3 — Two-Tier QR Security (Public vs. Verified Stay):** A printed QR code in Room 302 only identifies the room entry point. Anyone could photograph that QR code. Before showing the guest's bill or allowing room charges/orders, require Stay Verification (Guest enters Last Name + Room Number or SMS OTP). Issue a short-lived JWT (GuestAccessSession) that automatically revokes the moment the guest checks out or moves to another room.
- **Fix 4 — Exception States & Dual SLA Clocks:** Real hotel tasks don't just go from 'Received' to 'Completed'. Implement a strict backend state machine: Submitted -> Assigned -> Acknowledged -> In_Progress -> Completed -> Closed, plus Blocked (e.g. guest has Do Not Disturb on, or missing spare part), Cancelled, and Reopened. Track two separate SLA timers: Response SLA (time to acknowledge) and Resolution SLA (time to complete). Reassigning a ticket to another worker must NOT reset the SLA clock.

3. Core Project Modules: Simplified Feature Breakdown

To keep development clean and avoid scope creep, we removed low-ROI bloat (such as custom generative AI chatbots, complex points ledgers, voice transcription, and expensive smart-lock hardware replacements) and organized the core build into 6 modules:

Module	Key Features to Build	How It Works Technically (Implementation Summary)
1. Guest Portal (F01, F02, F05)	• Stay-linked QR login • Unified Stay Dashboard • Multi-language & WCAG UI	Mobile web app (React/Next.js). Public tier shows hotel info; private tier requires GuestAccessSession JWT. Uses i18n JSON files for instant language switching.
2. Booking & Pre-Arrival (F08, F09, F12, F14)	• Pre-arrival check-in link • Paid early/late checkout • Payment-PMS reconciliation	Sends automated link 24h before check-in. Checks live PMS room readiness before offering paid early check-in. Uses a transactional outbox table to sync payments with PMS.
3. Task Routing & SLAs (F15, F16, F17, F19)	• Auto-assignment by zone • Dual SLA timers • Shift handover sign-off	Router queries active staff by Property + Dept + Floor Zone + Lowest Task Count. Background worker checks SLA deadlines every 60s and fires supervisor escalation alerts.
4. Housekeeping & Eng. (F04, F21, F22, F25, F27)	• Room Readiness Board • Digital DND conflict guard • OOO Room Block approval	Separates Occupancy from Cleanliness (Dirty -> Cleaning -> Inspection_Pending -> Ready). Cleaners cannot mark 'Ready'—requires Supervisor tap, which updates PMS.
5. F&B & Services (F28, F29, F30, F31)	• Live menu & 86'd items • Allergen warning routing • Kitchen prep-time pacing	Syncs menu availability with POS. Uses client operation_id (idempotency key) so tapping 'Order' twice on slow Wi-Fi creates only 1 kitchen ticket.
6. Billing & Recovery (F32, F33, F34, F38, F41)	• Service recovery cases • Split billing & disputes • Checkout checklist & RBAC	Enforces role compensation caps (agents cannot self-approve above limit). Blocks digital checkout if disputed charges or unpaid balances exist. Enforces 4-check RBAC.

4. Suggested Additional Features & How to Build Them Simply

Beyond the baseline backlog, adding the following eight practical, low-complexity features will dramatically improve guest adoption, speed up payments, and protect frontline staff—without requiring heavy infrastructure or complex hardware:

Suggested Feature	Why Add It (Value)	How to Build It Simply (Technical Implementation)
1. WhatsApp Two-Way Guest Bridge	Most guests prefer texting on WhatsApp over opening browser tabs. Drives 2x higher engagement.	Use Meta WhatsApp Cloud API or Twilio. Map the guest's phone number from the PMS reservation to their active Stay_ID. Expose POST /api/webhooks/whatsapp. Convert quick-reply button taps (e.g. 'Need Towels') directly into tickets and stream free-text into the Front Desk inbox.
2. 1-Tap Instant Payment Links (UPI / Apple Pay / Google Pay)	Eliminates typing 16-digit card numbers for upgrades, F&B orders, or checkout settlement.	Integrate Razorpay Payment Links or Stripe Checkout Sessions. When a guest taps 'Pay Now', generate a dynamic payment link tied to stay_id and order_id. Process the payment.captured webhook idempotently via the IntegrationEvent table and post the receipt to the PMS folio.
3. Smart Mid-Stay Review Router (Day-2 Pulse Check)	Catches unhappy guests privately before checkout while routing happy guests to Google Reviews.	Run a scheduled background job (BullMQ / Cron) at 10:00 AM on Day 2 of the stay to send a 1-tap 1–5 star rating link. In the backend route: if rating <= 3, open a private feedback form and trigger a P0 RecoveryCase alert to the Duty Manager. If rating >= 4, redirect directly to the hotel's Google Review link.
4. Express Camera ID Scan & OCR Pre-Check-In	Saves 3–4 minutes at the Front Desk by auto-filling Passport / Government ID fields.	In the Pre-Arrival form (F08), use standard HTML5 camera capture (<input type='file' accept='image/*' capture='environment'>). Upload to a private S3 bucket with a 30-day auto-delete rule, and call AWS Textract or Google Vision OCR to auto-fill Name, DOB, and ID Number for staff review.

Suggested Feature	Why Add It (Value)	How to Build It Simply (Technical Implementation)
5. QR Minibar & Laundry Self-Declaration	Stops checkout arguments and chargebacks over minibar consumption.	Add a 'Minibar Consumed' quick-card on the Guest Dashboard (and on the cleaner's inspection screen). Tapping an item creates an idempotent StockMovement record (using a unique operation_id to prevent double-charging) and posts the charge to the PMS folio.
6. 1-Tap Staff SOS / Panic Alert Button	Protects housekeepers working alone in guest rooms; zero hardware cost.	Add a persistent red 'Hold 2s for SOS' button in the Staff PWA header. When triggered, read the staff member's active Task.room_id and broadcast an immediate WebSocket audio alarm + push notification to the Front Desk and Duty Manager consoles.
7. Automated Flight / Train Delay Pickup Sync	Prevents hotel drivers from waiting hours at the airport when a guest's flight is delayed.	Add an optional flight_number field to Airport Transfer requests (F31). A scheduled worker queries a Flight Status REST API (e.g. Aviationstack) 3 hours before pickup; if delayed >30 mins, it updates the task due time and alerts the Concierge queue.
8. Zero-Install PWA Push & Sound Alerts	Delivers native-style lock-screen alerts to staff and guests without App Store friction.	Add a manifest.json and Service Worker (sw.js) using the standard Web Push API (VAPID keys). Bind staff push subscriptions to their active Shift_ID upon login and auto-unsubscribe on clock-out.

5. Recommended Tech Stack & 3-Sprint Build Sequence

The tech team can build and ship the entire platform rapidly using a clean, standard web stack:

- **Frontend (Guest Portal, Staff PWA, Admin Console):** Next.js / React (TypeScript) configured as a Progressive Web App (PWA) with Tailwind CSS for high-contrast, mobile-first screens.
- **Backend API & Real-Time Layer:** Node.js (NestJS / Express) or Python (FastAPI) with REST endpoints + Socket.io / WebSockets for live task boards and SLA countdowns.
- **Database & Background Jobs:** PostgreSQL (relational integrity for stays, tasks, and audit logs) + Redis & BullMQ (for SLA timers, Day-2 pulse checks, and webhook retry queues).

- **Storage & Integration Adapters:** Private S3-compatible bucket (for maintenance photos, Lost & Found images, and temporary ID scans) + Adapter Pattern interfaces for PMS/POS/Payment APIs (using clean Mock Adapters in dev/staging).

Suggested Implementation Sprints

- **Sprint 1 (Weeks 1–4) — Core Foundation & State Machines:** Build the database schema, 4-check RBAC middleware, Stay-Linked QR verification (F01), Guest Dashboard (F02), Ticket State Machine & Dual SLA clocks (F16/F17), Room Readiness Board (F21), Shift Handover (F19), and Mock PMS/Payment adapters.
- **Sprint 2 (Weeks 5–8) — Live Integrations, Billing & F&B:** Implement the Transactional Webhook Outbox, Payment-Booking Reconciliation (F14), Live Menu & F&B Ordering (F28/F29), Digital Checkout Checklist (F34), Service Recovery Cases (F32), and Instant Payment Links (NEW-02).
- **Sprint 3 (Weeks 9–12) — Automation, Upsells & High-Value Additions:** Add Pre-Arrival OCR Registration (F08 / NEW-04), Paid Early/Late Checkout & Upgrades (F09/F35), Digital DND Sync (F04), Preventive Maintenance & Room Blocks (F25/F27), WhatsApp Bridge (NEW-01), and the Day-2 Review Router (NEW-03).

6. Developer Starter Cheat Sheet: Key API Routes & Tables

Use this starter endpoint and table mapping to scaffold the backend API and PostgreSQL schema:

REST Endpoint	Allowed Roles	Core Action & Idempotency / Validation Rule
POST /api/stay/verify	Public (QR Scan)	Matches Room # + Last Name/OTP against PMS; issues stay-scoped JWT.
POST /api/requests	Guest (Verified)	Creates request + auto-routes Task; deduplicates by client operation_id.
PATCH /api/tasks/:id/state	Staff / Supervisor	Validates state transition; logs StatusEvent; pauses SLA only if approved.
PATCH /api/rooms/:id/ready	Supervisor Only	Requires completed checklist; marks room Ready and pushes status to PMS.
POST /api/webhooks/:provider	Signed Webhook	Verifies HMAC signature; saves to IntegrationEvent outbox before processing.
POST /api/stay/checkout	Guest / Front Desk	Verifies $0 balance & 0 open FolioDisputes; triggers PMS checkout & revokes JWT.